Gastro-oesophageal Reflux Disease (GERD): Approach to a Case of Heartburn

Dr. Muhammad Reihan, Associate Professor of Medicine | Source: `GERD.pdf` (25 slides)

Legend: ⭐ = high-yield · *[added]* = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes. The lecture has **no answer key** for its case; the answer below is **[reasoning added; key not on slides]**.

🔗 **Bridge from earlier lectures:**

- **CAD / ECG (Reihan, same lecturer):** retrosternal burning can be **ACS**. Slide 14: "**rule out life-threatening causes (ACS) first**". *[added]* ECG + troponin before labelling it reflux. **Nitrates lower LES tone** (slide 8): the anti-anginal itself can cause heartburn.
- **Asthma / COPD (Kotb):** GERD is both a **risk factor** and a **complication**:
- **asthma** and **chronic cough** (nocturnal)
- **chronic bronchitis**
- **aspiration pneumonia** (link: Pneumonia lecture)
- **Restrictive Lung Disease (Kotb): scleroderma** → oesophageal dysmotility → severe GERD; *[added]* reflux/micro-aspiration is linked to **ILD/IPF**.
- **Haematology 1: iron-deficiency anaemia** from oesophageal erosions. IDA in GERD is a **red flag** → EGD.
- **Gastric Tumours:**
- adenocarcinoma of the **distal oesophagus/GEJ** arises from **Barrett**; *[added]* the squamous cell type is linked to smoking/alcohol
- **H. pylori triple therapy** (case option D) treats peptic ulcer, not Barrett

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1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	⭐ Mechanisms and risk factors: **GEJ dysfunction** and **impaired acid clearance**	Transient LES relaxations; nitrates/caffeine; obesity, pregnancy; smoking ↓ saliva; hiatal hernia
Knowledge	Typical, atypical and **extra-oesophageal** manifestations	Cough, hoarseness, asthma, aspiration
Skill	⭐ Stepwise diagnosis: **empiric PPI trial vs upfront EGD**	Red flags; Barrett risk factors; pH monitoring not routine
Skill	Management: **lifestyle, PPI, surgical referral**	Nissen indications; Roux-en-Y in obesity; Barrett surveillance/ablation

Main subtopics (slide 4):

- definition and pathophysiology
- aetiology and risk factors
- clinical features
- differential diagnosis
- diagnostics

- treatment
- **Barrett oesophagus**
- complications

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2. Main Content (Lecture Order)

2.0 Opening Case (Slides 5–6)

Presentation:

- **58-year-old man**, **12 years of recurrent heartburn**
- **nocturnal cough** for 1 year (extra-oesophageal GERD)
- not seen a doctor for 8 years
- **30 pack-years**; OTC antacid only
- **BMI 31** (obese); exam and CBC normal
- **EGD: dark-pink (salmon) mucosa extending 2 cm above the GEJ**
- **biopsy: columnar epithelium with goblet cells** + ⭐ **low-grade dysplasia**

Next step? → ⭐ **B. Endoscopic radiofrequency ablation** *[reasoning added; key not on slides]*:

- columnar epithelium + goblet cells = **Barrett oesophagus** (intestinal metaplasia)
- slide 20: for **low-grade dysplasia**, **endoscopic eradication** is first; **surveillance every 12 months** is an acceptable alternative
- *[added]* LGD must first be **confirmed by a second, expert GI pathologist**; give **high-dose PPI** alongside

Why not the others:

- **A (repeat EGD in 24 months):** the interval is too long. Even the LGD surveillance alternative is **12 months**. *[added]* 3–5 years is for non-dysplastic Barrett.
- **C (oesophagectomy):** for **invasive cancer**; overtreatment for LGD
- **D (PPI + clarithromycin + metronidazole):** *H. pylori* eradication; no role in Barrett
- **E (radiotherapy):** for cancer
- **F (Nissen fundoplication):** controls reflux but **does not reverse metaplasia or eliminate dysplasia**

2.1 Definitions (Slide 7) ⭐

Term	Definition
Gastro-oesophageal reflux	Regurgitation of stomach contents into the oesophagus (can be physiological)
⭐ **GERD**	Reflux causing **troublesome symptoms** (typically **heartburn or regurgitation**) **and/or oesophageal injury/complications**

2.2 Aetiology: Mechanisms (Slide 8) ⭐⭐

A. Gastro-oesophageal junction dysfunction:

1. ⭐ ↑ **frequency of transient LES relaxations (TLESRs)** (*[added]* the **main** mechanism; not swallow-related)
2. **imbalance between intragastric and LES pressures:**

- ↓ **LES tone**: **caffeine, nitroglycerin** (*[added]* also CCBs, anticholinergics, theophylline, progesterone, alcohol, chocolate, peppermint, fatty food)
- ↑ **intragastric pressure**: **pregnancy, delayed gastric emptying, obesity**

3. **anatomical GEJ abnormalities** (*[added]* **hiatal hernia** displaces the LES above the crural diaphragm)

B. Impaired oesophageal acid clearance:

- ↓ **salivation** (e.g. **smoking**); saliva bicarbonate neutralises acid
- ↓ **peristalsis** (*[added]* scleroderma, diabetes)

2.3 Risk Factors (Slide 9) ⭐

Category	Risk factors
Lifestyle	**Smoking, caffeine, alcohol, stress**
↑ Abdominal pressure	**Obesity, pregnancy**
Motility/connective tissue	⭐ **Scleroderma** (*[added]* smooth-muscle atrophy → aperistalsis + incompetent LES)
Anatomy	⭐ **Sliding hiatal hernia**, present in **≥ 90 % of patients with severe GERD**
Respiratory	**Asthma** (bidirectional: *[added]* bronchodilators lower LES tone; hyperinflation alters the crura)

2.4 Clinical Features (Slides 10–12) ⭐⭐

Typical	Atypical	Extra-oesophageal
⭐ **Heartburn** (retrosternal burning)	**Chest pressure** (mimics angina)	⭐ **Chronic non-productive cough**
⭐ **Regurgitation**	**Belching**	⭐ **Hoarseness** (reflux laryngitis)
Dysphagia	**Aspiration pneumonia** features	*[added]* asthma, dental erosion, globus, sinusitis

Aggravating factors:

- **lying down shortly after meals**
- certain foods and drinks
- *[added]* bending, large/fatty meals

Red flags in GERD (slide 12):

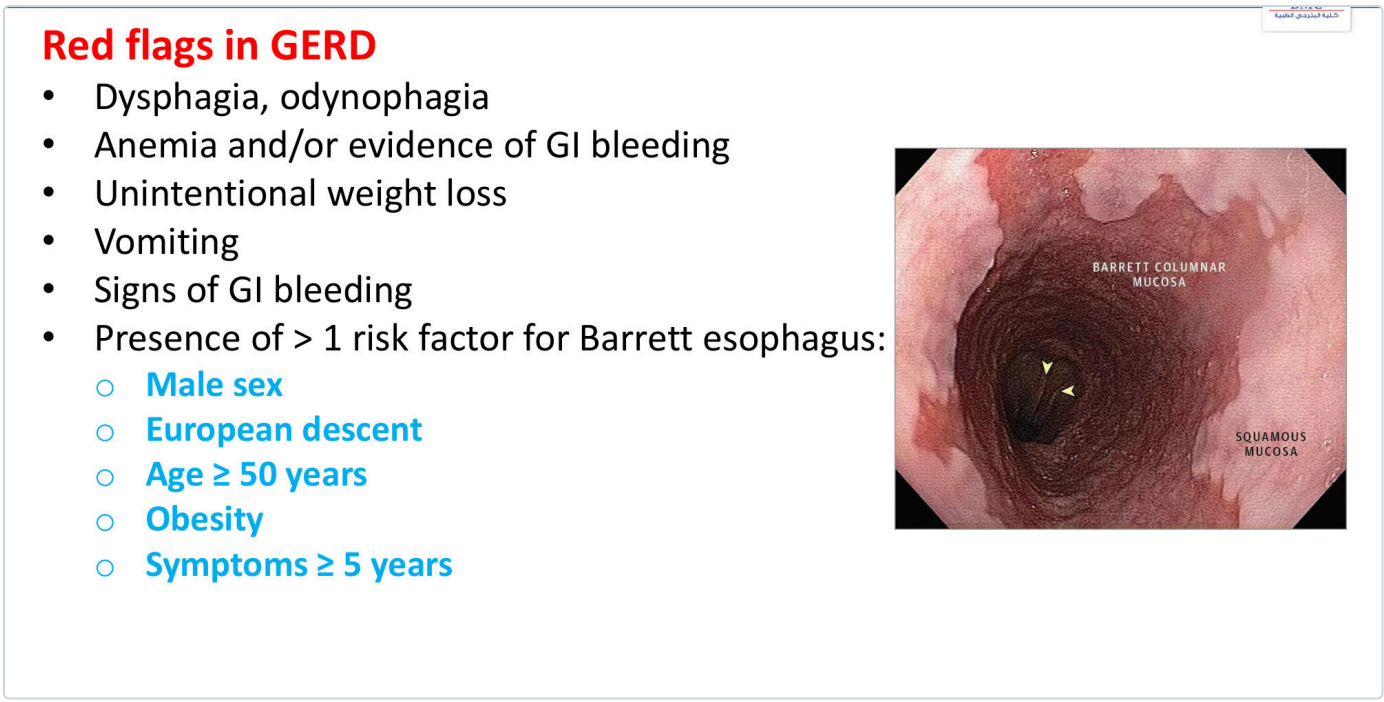
Red flag
⭐ **Dysphagia, odynophagia**
⭐ **Anaemia and/or GI bleeding**
⭐ **Unintentional weight loss**
Vomiting
> 1 risk factor for Barrett oesophagus (below)

[added] Slide 12 lists "GI bleeding" twice.

Barrett risk factors (slide 12; repeated slide 18):

- **male sex**
- **European descent (White)**

- **age ≥ 50**
- **obesity**
- **symptoms ≥ 5 years**
- *[added]* also smoking, family history of Barrett/oesophageal adenocarcinoma, hiatal hernia; ACG uses **≥ 3 risk factors** for screening



Slide 12: Endoscopy: Barrett columnar mucosa vs squamous mucosa

Mnemonic for the Barrett risk factors, "OLD, WHITE, OBESE MALE with a LONG HISTORY": age ≥ 50 · White/European · obesity · male · symptoms ≥ 5 years.

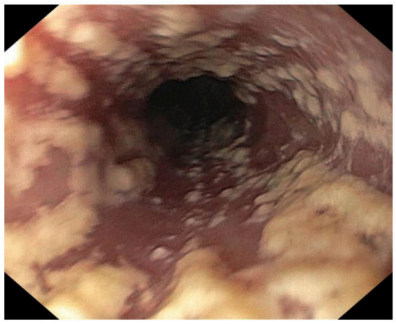
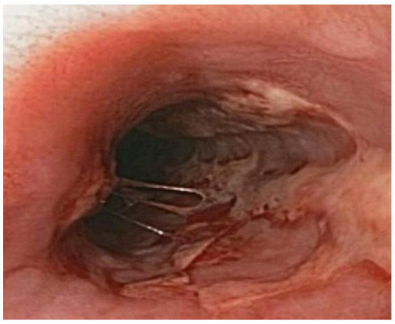
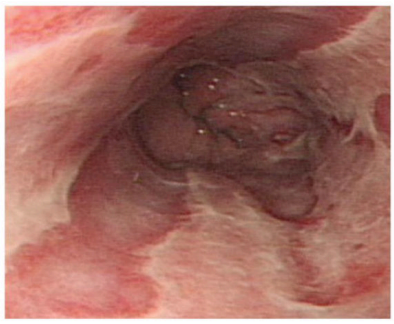
2.5 Differential Diagnosis (Slide 13) ⭐

Infectious oesophagitis (typically **immunocompromised**):

Organism	Endoscopy
Candida	⭐ **White/yellow adherent plaques (pseudomembranes)**
HSV (mainly HSV-1)	⭐ **Punched-out** (small, "volcano") ulcers
CMV	⭐ **Linear** (large, shallow, longitudinal) ulcers

Esophagitis

- Infectious esophagitis (typically seen in immunocompromised patients)
  - Esophageal candidiasis: white or yellow adherent plaques (pseudomembranes).
  - Herpes esophagitis: (mainly HSV-1); punched-out ulcers
  - CMV esophagitis: linear ulcers

Slide 13: Candida plaques, HSV punched-out ulcers, CMV linear ulcers

[added] **Other DDs not on the slide:**

- **ACS** (slide 14 says rule out first)
- **eosinophilic oesophagitis** (young atopic man, food impaction, **rings/furrows**, PPI-resistant)
- **pill oesophagitis** (doxycycline, NSAIDs, bisphosphonates, KCl)
- **peptic ulcer disease**, **achalasia**, **oesophageal cancer**, **functional heartburn**

2.6 Diagnostics (Slides 14–15) ⭐⭐⭐

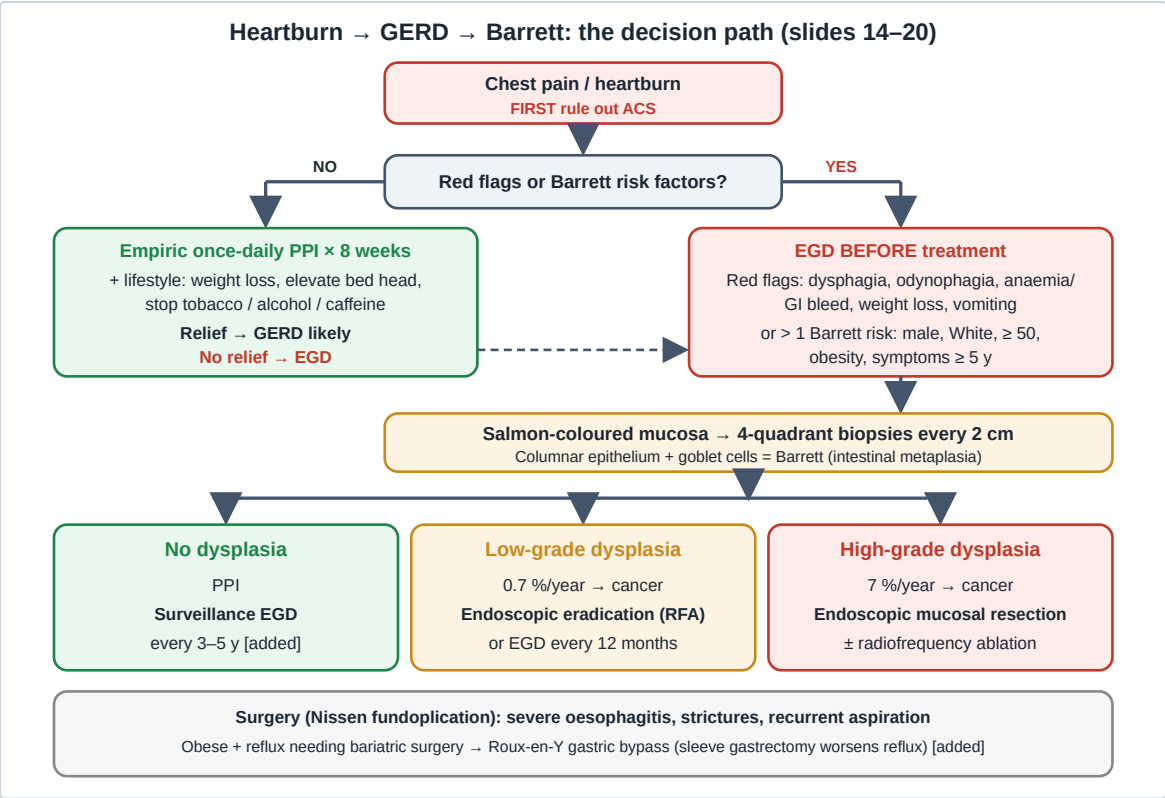
Approach (slide 14):

1. **All patients: ⭐ rule out life-threatening causes of chest pain (ACS)** first
2. **Typical symptoms without red flags:**
   - treat as GERD: ⭐ **empiric once-daily PPI trial**
   - **relief after 8 weeks of PPI → GERD likely**
3. **Red flags** (slide 15): ⭐ **refer for EGD before starting treatment**

⭐ EGD indications (slide 15)
Red flags
Risk factors for Barrett oesophagus
No improvement after the PPI trial

Oesophageal pH monitoring:

- ⭐ **not a routine diagnostic test**
- *[added]* used for **PPI-refractory** symptoms with a normal EGD, **before anti-reflux surgery**, or when the diagnosis is uncertain; off PPI, 24-h catheter or 96-h wireless capsule
- *[added]* **Manometry** before surgery excludes achalasia/aperistalsis



Added diagram: Heartburn → GERD → Barrett decision path

2.7 Treatment (Slides 16–17) ⭐⭐

Pharmacological:

- ⭐ **PPI at standard dose** (*[added]* omeprazole 20 mg, 30–60 min before breakfast, for 8 weeks; then the lowest effective dose / on demand)
- *[added]* step up to **twice daily** if partial response
- *[added]* **H2RAs / alginates** for breakthrough nocturnal symptoms
- *[added]* **long-term PPI risks**: B12, Mg^{2+} and iron deficiency, *C. difficile*, fractures (link: Haematology 2)

Lifestyle:

- "physical recommendations"
- ⭐ **weight loss in obesity** (*[added]* most effective lifestyle measure)
- ⭐ **elevate the head of the bed** (*[added]* for nocturnal symptoms; avoid eating 2–3 h before bed)
- **avoid triggers**: **tobacco, alcohol, caffeine**

Surgical therapy (slide 17):

- **Fundoplication** indications (slide):
  - ⭐ **severe oesophagitis**
  - **strictures**
  - **recurrent aspiration**
  - *[added]* also PPI-intolerant, large hiatal hernia, regurgitation-dominant symptoms, patient preference
- **Technique: ⭐ complete (360°) Nissen fundoplication**: the gastric fundus is wrapped around the lower oesophagus

- **Obesity + reflux needing bariatric surgery:** ⭐ **Roux-en-Y gastric bypass** (*[added]* sleeve gastrectomy worsens reflux)

Surgical therapy

Fundoplication

Indications

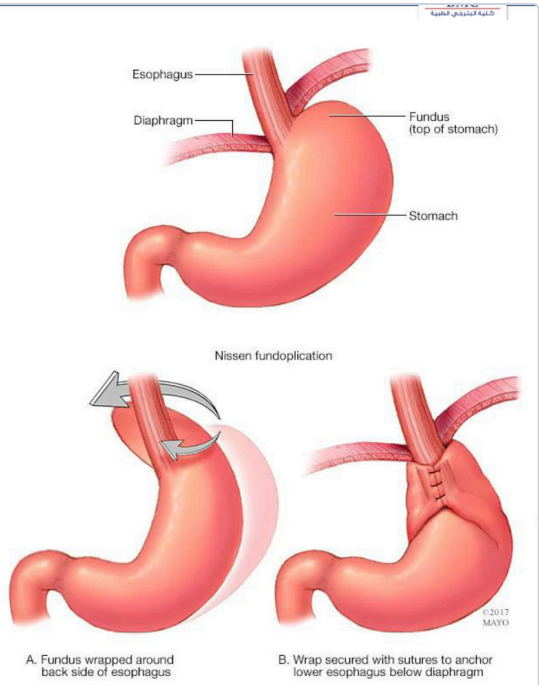
severe esophagitis, strictures, recurrent aspiration

Techniques

Complete fundoplication (Nissen fundoplication)

Considerations for patients with comorbidities

Patients with obesity and reflux undergoing bariatric surgery: Consider Roux-en-Y.



Slide 17: Nissen fundoplication

> 🔁 **Recap: GERD.**
>
> - **Mechanisms:** TLESRs, ↓ LES tone (caffeine, nitrates), ↑ gastric pressure (obesity, pregnancy), **hiatal hernia**, ↓ clearance (smoking, dysmotility).
> - **Symptoms:** heartburn, regurgitation, dysphagia; cough, hoarseness.
> - **Diagnosis:** rule out ACS → no red flags = **8-week PPI trial**; red flags or Barrett risk = **EGD first**; pH study not routine.
> - **Treatment:** PPI + weight loss + bed elevation; **Nissen** for severe oesophagitis, stricture, aspiration; **Roux-en-Y** if obese.
>
> 🔗 **Link:** years of acid injury drive the key premalignant complication.
>
> ▶️ **Up next: Barrett oesophagus.**

2.8 Barrett Oesophagus (Slides 18–20) ⭐⭐⭐

Definition:

- ⭐ **intestinal metaplasia** of the oesophageal mucosa caused by **chronic reflux**
- **columnar** epithelium replaces the normal **squamous** epithelium
- a **premalignant** change

Pathophysiology (slide 19):

1. reflux oesophagitis: acid damages the distal oesophageal mucosa
2. **non-keratinised stratified squamous** epithelium is replaced by
3. **non-ciliated columnar epithelium with goblet cells** = **intestinal (Barrett) metaplasia**

⭐ **Complication: oesophageal adenocarcinoma** (distal third / GEJ).

Barrett esophagus

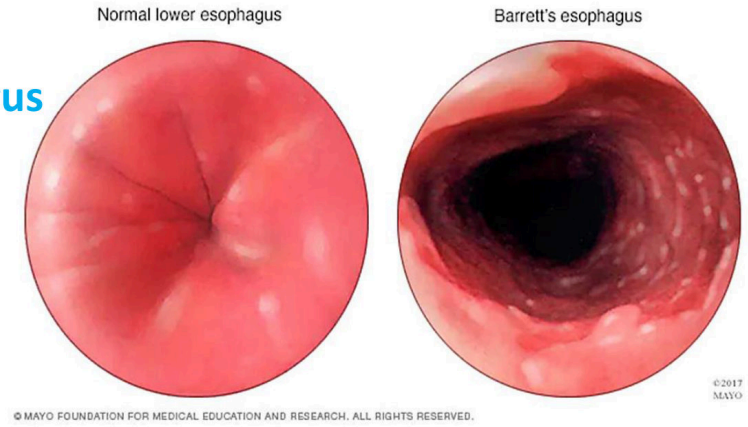
Definition:

Intestinal metaplasia of the esophageal mucosa induced by chronic reflux.

- Columnar epithelium instead of the normal squamous epithelium
- A premalignant change

Risk factors for Barrett esophagus

- Male sex
- European descent
- Age ≥ 50 years
- Obesity
- Symptoms ≥ 5 years



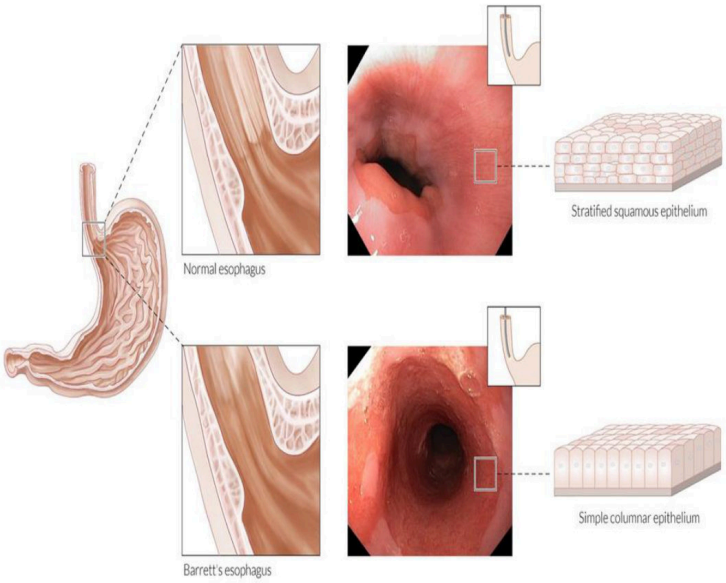
Slide 18: Normal lower oesophagus vs Barrett oesophagus

Barrett esophagus

Pathophysiology

Reflux esophagitis
→ stomach acid damages mucosa of distal esophagus
→ nonkeratinized stratified squamous epithelium
→nonciliated columnar epithelium and goblet cells
= (intestinal metaplasia, Barrett metaplasia)

Complications: esophageal adenocarcinoma



Slide 19: Squamous → columnar metaplasia

Management and surveillance (slide 20):

- **Endoscopy** with **four-quadrant biopsies every 2 cm** of the suspicious **salmon-coloured** mucosa (*[added]* Seattle protocol)

Histology	Malignant transformation	Management
[added] **No dysplasia**	~0.2–0.5 %/yr	PPI; surveillance EGD every **3–5 years**
⭐ **Low-grade dysplasia**	⭐ **0.7 % per year**	**Endoscopic eradication** (RFA); **or surveillance EGD every 12 months** (acceptable alternative)

Histology	Malignant transformation	Management
⭐ **High-grade dysplasia**	⭐ **7 % per year**	**Endoscopic treatment of mucosal irregularities**; **eradication of dysplasia** via **endoscopic mucosal resection** and/or **radiofrequency ablation**

[added] Oesophagectomy is reserved for **invasive** cancer beyond the mucosa. **Fundoplication does not cause Barrett regression**.

Mnemonic for the transformation rates, "LOW = 0.7, HIGH = 7": move the decimal point one place: low-grade 0.7 %/yr, high-grade 7 %/yr.

2.9 Other Complications (Slides 21–23) ⭐⭐

Iron deficiency anemia:
Mucosal erosions and ulcerations → chronic bleeding → anemia

Esophageal stricture
Most common sequela of reflux esophagitis
Diagnostics
- Barium esophagram (best initial test)
- Endoscopy with biopsies: to rule out malignancy

Treatment
Dilation with bougie dilator/balloon dilator

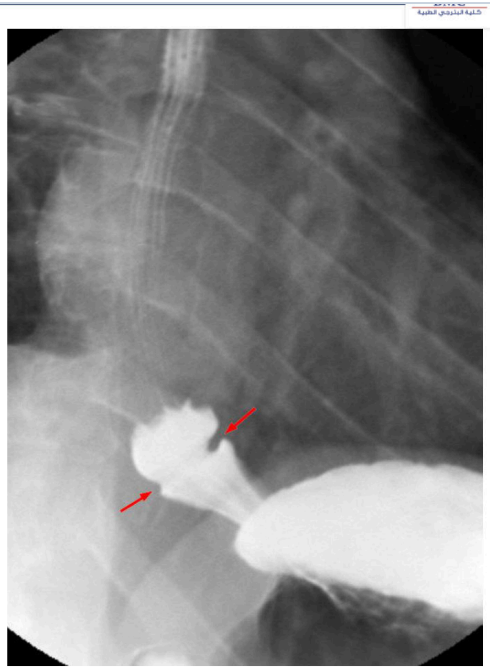


Slide 21: Barium swallow: peptic stricture

Complication	Key points
Iron-deficiency anaemia	Mucosal **erosions/ulcers → chronic bleeding** → IDA
⭐ **Oesophageal (peptic) stricture**	**Commonest sequela of reflux oesophagitis**. **Diagnosis:** ⭐ **barium oesophagram (best initial test)**; then **EGD with biopsies to rule out malignancy**. **Treatment: dilation (bougie or balloon)** (*[added]* + long-term PPI)
⭐ **Schatzki ring**	Thin narrowing ring, **most often at the squamocolumnar junction**; **usually caused by chronic reflux**; causes **dysphagia** (*[added]* intermittent solid-food impaction, "steakhouse syndrome"); **treated with dilation**
Aspiration complications	**Aspiration pneumonia**, **chronic bronchitis**, **asthma**, **reflux laryngitis (hoarseness)**

Esophageal ring

- Schatzki rings: narrowing of the esophagus
  - Most commonly seen at the squamocolumnar junction
  - Usually caused by chronic acid reflux
  - Can lead to dysphagia
  - Usually treated with esophageal dilation



Slide 22: Schatzki ring on barium swallow

[added] **Dysphagia pattern clues:**

- **progressive solids-only** → stricture or cancer
- **intermittent solids** → ring/web
- **solids + liquids from onset** → motility disorder (achalasia, scleroderma)

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3. High-Yield Comparison Tables

Table A: Empiric PPI vs EGD First

[slides 14–15]

Empiric PPI trial (8 weeks)	EGD before treatment
Typical heartburn/regurgitation	**Red flags**: dysphagia, odynophagia, anaemia, GI bleeding, weight loss, vomiting
No red flags	**> 1 Barrett risk factor**: male, White, ≥ 50, obese, symptoms ≥ 5 y
≤ 1 Barrett risk factor	**No response to PPI**

Table B: Infectious Oesophagitis

	Candida	**HSV**	**CMV**
Endoscopy	**White/yellow plaques** (pseudomembranes)	**Punched-out** small ulcers	**Linear**, large shallow ulcers
Host	Immunocompromised, inhaled steroids *[added]*	Immunocompromised	Immunocompromised (AIDS, transplant)
Treatment *[added]*	Fluconazole	Aciclovir	Ganciclovir

Table C: Barrett Management by Grade

Grade	Cancer risk/yr	Action
No dysplasia *[added]*	~0.3 %	PPI; EGD every 3–5 y
LGD	**0.7 %**	**RFA** (or EGD every **12 months**)
HGD	**7 %**	**EMR ± RFA**
Invasive cancer *[added]*	—	Oesophagectomy / oncology

Table D: Surgical Options

Situation	Operation
Severe oesophagitis, stricture, recurrent aspiration	**Nissen (360°) fundoplication**
Obese + reflux, bariatric candidate	**Roux-en-Y gastric bypass**

Table E: Stricture vs Schatzki Ring

	Peptic stricture	Schatzki ring
Cause	Chronic reflux oesophagitis	Chronic reflux (usually)
Site	Distal oesophagus	**Squamocolumnar junction**
Dysphagia *[added]*	**Progressive**, solids	**Intermittent**, solids (food bolus)
First test	**Barium oesophagram** → EGD + biopsy	Barium / EGD
Treatment	**Bougie/balloon dilation**	**Dilation**

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4. Rules, Exceptions, Patterns

Rule	Exception / nuance
Heartburn = GERD → PPI	**Rule out ACS first** (slide 14)
Empiric PPI trial	**Not if red flags or > 1 Barrett risk factor** → EGD first
pH monitoring confirms GERD	**Not routine**; *[added]* for refractory symptoms / pre-surgery
Fundoplication cures Barrett	❌ Controls reflux; **dysplasia still needs endoscopic eradication/surveillance**
Barrett LGD → watch	**Eradication (RFA) preferred**; 12-monthly surveillance is the acceptable alternative
Obese GERD patient → sleeve gastrectomy	❌ *[added]* Sleeve worsens reflux → **Roux-en-Y** (slide 17)
Nitrates for chest pain	**Lower LES tone** → can worsen reflux
Smoking only affects the lungs	↓ saliva → **impaired acid clearance**; Barrett risk factor *[added]*
Stricture → endoscopy first	**Barium oesophagram is the best initial test** (slide 21), then EGD + biopsy
Asthma and GERD	**Bidirectional**: asthma is a risk factor (slide 9) and a complication (slide 23)

Pattern recognition:

- **heartburn + nocturnal cough + hoarseness** → extra-oesophageal GERD
- **> 50 y, White, obese, male, long reflux + salmon mucosa + goblet cells** → **Barrett**

- **long GERD + progressive solid dysphagia, no weight loss** → **peptic stricture** (weight loss → cancer)
- **intermittent food sticking at a meal** → **Schatzki ring**
- **immunocompromised + odynophagia + white plaques** → **Candida**
- **scleroderma + severe GERD** → aperistalsis + LES failure

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5. Clinical Correlations

- **Chest pain in the ED:** ECG/troponin first. "GI cocktail" relief does **not** exclude ACS *[added]*.
- **Patients on nitrates/CCBs** (CAD/HTN lectures) may develop or worsen reflux. Consider this before escalating PPIs.
- **Chronic cough with a normal CXR: GERD, asthma, upper-airway cough (post-nasal drip)** *[added]* are the big three; plus ACE inhibitors.
- **Obesity:**
  - **weight loss** is the most effective lifestyle measure
  - bariatric surgery → **Roux-en-Y** for refluxers
- **IDA in a GERD patient** → EGD (red flag); also look for other GI sources (link: Haematology 1).
- **Long-term PPIs** *[added]*: review the indication. Risks include B12 and Mg^{2+} deficiency, *C. difficile*, fractures. Barrett patients **do** need long-term PPI.
- **Barrett screening:** men with chronic GERD plus other risk factors (≥ 50, White, obese, smoker, family history).

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6. Rapid Revision (Last-Day)

- **Definition:** GERD = reflux causing **troublesome symptoms** (heartburn/regurgitation) **and/or injury/complications**.
- **Mechanisms:**
  - **GEJ dysfunction**: ↑ **transient LES relaxations**; ↓ LES tone (**caffeine, nitroglycerin**); ↑ intragastric pressure (**pregnancy, delayed gastric emptying, obesity**); anatomical (hiatal hernia)
  - **↓ acid clearance**: **↓ saliva (smoking)**, **↓ peristalsis**
- **Risk factors: smoking, caffeine, alcohol, stress, obesity, pregnancy, scleroderma, sliding hiatal hernia (≥ 90 % of severe GERD), asthma**.
- **Symptoms:**
  - **typical**: heartburn, regurgitation, dysphagia
  - **atypical**: chest pressure, belching, aspiration pneumonia
  - **extra-oesophageal**: **chronic dry cough, hoarseness**
  - worse **lying down after meals**
- **Red flags: dysphagia/odynophagia, anaemia/GI bleed, weight loss, vomiting, > 1 Barrett risk factor**.
- **Barrett risk factors: male, European/White, ≥ 50, obesity, symptoms ≥ 5 years**.
- **DDx: infectious oesophagitis:**
  - ***Candida***: **white/yellow plaques**
  - **HSV-1**: **punched-out ulcers**
  - **CMV**: **linear ulcers**
- **Diagnosis:**
  - **rule out ACS** → no red flags: **once-daily PPI trial; relief at 8 weeks = GERD likely**

- red flags → **EGD before treatment**
- **EGD indications**: red flags, Barrett risk factors, **PPI failure**
- **pH monitoring not routine**

- **Treatment:**
  - **standard-dose PPI**
  - **weight loss**, **elevate head of bed**, avoid **tobacco, alcohol, caffeine**
- **Surgery:**
  - **Nissen (complete) fundoplication** for **severe oesophagitis, strictures, recurrent aspiration**
  - obese + bariatric → **Roux-en-Y**
- **Barrett oesophagus:**
  - **intestinal metaplasia**: squamous → **non-ciliated columnar + goblet cells**; premalignant → **adenocarcinoma**
  - surveillance: **4-quadrant biopsies every 2 cm** of salmon mucosa
  - **HGD 7 %/yr → EMR ± RFA**
  - **LGD 0.7 %/yr → endoscopic eradication**, or **EGD every 12 months**
- **Other complications:**
  - **IDA** (erosions)
  - **stricture**: commonest sequela; **barium swallow first**, EGD + biopsy, **bougie/balloon dilation**
  - **Schatzki ring**: at the **squamocolumnar junction**, reflux, dysphagia, **dilation**
  - **aspiration**: pneumonia, chronic bronchitis, asthma, **reflux laryngitis**

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7. Exam-Style Questions

Q1. (Lecture case) A 58-year-old obese smoker has 12 years of heartburn and a nocturnal cough. EGD shows salmon-pink mucosa 2 cm above the GEJ; biopsy shows **columnar epithelium with goblet cells and low-grade dysplasia**. Next step? A. Repeat EGD in 24 months · **B. Endoscopic radiofrequency ablation** · C. Oesophagectomy · D. Omeprazole + clarithromycin + metronidazole · E. External-beam radiotherapy · F. Nissen fundoplication

> **Answer: B** *[reasoning added; key not on slides]*.
> - **Barrett with LGD → endoscopic eradication** (slide 20). The acceptable alternative is EGD in **12**, not 24, months.
> - C/E are for cancer. D is *H. pylori* therapy. F doesn't eliminate dysplasia.
> - *[added]* Confirm LGD with an expert pathologist; continue PPI.

Q2. A 32-year-old woman has 3 months of postprandial heartburn and regurgitation, no weight loss, dysphagia, anaemia or vomiting, and a normal ECG. Best next step? A. Urgent EGD · **B. Empiric once-daily PPI for 8 weeks + lifestyle advice** · C. 24-h pH monitoring · D. Barium swallow · E. Nissen fundoplication

> **Answer: B** (slide 14). Typical symptoms, no red flags, < 2 Barrett risk factors. Relief at 8 weeks → GERD likely.

Q3. Which patient needs **EGD before** starting a PPI? A. 28-year-old with heartburn after spicy meals · **B. 55-year-old man with heartburn and new progressive dysphagia to solids** · C. 35-year-old

pregnant woman with heartburn · D. 40-year-old with nocturnal regurgitation relieved by antacids · E. 25-year-old with belching

> **Answer: B** (slides 12, 15). **Dysphagia** is a red flag, and he has > 1 Barrett risk factor (male, ≥ 50).

Q4. A patient with long-standing GERD has slowly progressive dysphagia to solids without weight loss. Best **initial** test? **A. Barium oesophagram** · B. CT chest · C. Manometry · D. pH monitoring · E. Chest X-ray

> **Answer: A** (slide 21).
> - Suspected **peptic stricture**: barium oesophagram first, then **EGD + biopsy** to exclude malignancy, then **dilation**.
> - *[added]* Many guidelines go straight to EGD; for this lecture, barium first.

Q5. An HIV-positive patient (CD4 50) has severe odynophagia. EGD shows **long, linear, shallow ulcers** in the distal oesophagus. Most likely organism? A. *Candida* · B. HSV-1 · **C. CMV** · D. *H. pylori* · E. EBV

> **Answer: C** (slide 13).
> - **Linear ulcers = CMV**; punched-out = HSV; plaques = *Candida*.
> - *[added]* Treatment: ganciclovir.

Q6. An obese patient (BMI 42) with severe, PPI-dependent GERD wants bariatric surgery. Which procedure does the lecture recommend? A. Sleeve gastrectomy · B. Gastric band · **C. Roux-en-Y gastric bypass** · D. Nissen fundoplication alone · E. Biliopancreatic diversion

> **Answer: C** (slide 17). It treats obesity **and** reflux. *[added]* Sleeve gastrectomy worsens GERD.

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8. Exam Pitfalls & Traps

Trap	Correct thinking
Heartburn → PPI straight away	❌ **Rule out ACS first**
Empiric PPI for everyone	❌ **Red flags or > 1 Barrett risk factor → EGD before treatment**
pH monitoring to diagnose GERD	❌ **Not routine**
Barrett LGD → repeat EGD in 2 years	❌ **RFA**, or EGD in **12 months**
Barrett HGD → oesophagectomy	❌ **Endoscopic resection ± RFA**; surgery is for invasive cancer *[added]*
Fundoplication reverses Barrett	❌ It controls reflux only; dysplasia management is unchanged
H. pylori eradication for Barrett	❌ Irrelevant (*[added] H. pylori* may even be inversely associated)
Barrett = squamous metaplasia	❌ **Squamous → columnar (with goblet cells)** = intestinal metaplasia
Barrett → squamous cell carcinoma	❌ **Adenocarcinoma**
HGD 0.7 %, LGD 7 %	❌ Reversed: **LGD 0.7 %/yr, HGD 7 %/yr**
Obese GERD → sleeve gastrectomy	❌ **Roux-en-Y**
Stricture → EGD as the first test	⚠️ Lecture: **barium oesophagram first**, then EGD + biopsy
HSV → linear ulcers	❌ **HSV punched-out; CMV linear**

Trap	Correct thinking
Nitrates are safe in GERD	⚠️ **Lower LES tone**
Chronic cough is pulmonary	⚠️ GERD causes **chronic non-productive cough** and **hoarseness**